

Project Title: AI Based Fitness & Nutrition Assistant

Introduction:

Project Overview:

AI-Based Fitness & Nutrition Assistant is a web-based platform designed to help beginners, students, and budget-conscious users maintain a healthier lifestyle through personalized workout and nutrition guidance. Many users face difficulty choosing a suitable workout or diet plan because their body condition, fitness goal, lifestyle, and budget are different.

This project will allow users to register, log in, create a fitness profile, receive workout and diet recommendations, view exercise guidance, and track their progress. Gemini API will be used to generate AI-based workout suggestions, nutrition guidance, progress analysis, and motivational recommendations.

Objectives:

- The main objectives of this project are:
- To provide personalized workout guidance based on user profile and fitness goal.
- To suggest affordable and locally suitable nutrition options.
- To help users track their weight and progress.
- To use AI for adaptive fitness and nutrition recommendations.
- To make fitness guidance simple and beginner-friendly.

Scope:

The scope of this project is to develop a web-based fitness and nutrition assistant with authentication, fitness profile management, workout recommendation, nutrition recommendation, exercise library, progress tracking, dashboard, and admin content management.

The system will focus on general fitness guidance, local affordability, and beginner accessibility. It will not replace professional doctors, certified nutritionists, or personal trainers.

Functional Requirements:

- **FR-01:** Users can register using name, email, and password.

- **FR-02:** Users can log in securely using email and password.
- **FR-03:** The system can protect user accounts using password hashing and JWT authentication.
- **FR-04:** Users can create and update their fitness profile with age, gender, height, weight, activity level, fitness goal, and budget preference.
- **FR-05:** The system can calculate basic fitness data such as BMI from the user profile.
- **FR-06:** Users can receive personalized workout recommendations based on their profile and fitness goal.
- **FR-07:** Users can receive budget-based nutrition recommendations suitable for local food availability.
- **FR-08:** Gemini AI can provide workout guidance, nutrition suggestions, progress feedback, and motivational recommendations.
- **FR-09:** Users can view an exercise library with exercise descriptions, difficulty level, target body part, and video guidance.
- **FR-10:** Users can track their weight and progress history.
- **FR-11:** Users can view profile summary, BMI, progress overview, and recommendations from the dashboard.
- **FR-12:** Admin can manage exercise information and exercise categories.
- **FR-13:** The system can store user data, profile data, progress data, exercise data, and recommendation history in the database.
- **FR-14:** The frontend can communicate with the backend through REST APIs and display proper success or error messages.

Non-Functional Requirements;

- **NFR-01:** The system should have a simple and beginner-friendly user interface.
- **NFR-02:** The system should be responsive and usable on desktop, tablet, and mobile devices.
- **NFR-03:** User passwords must be securely hashed before storing.
- **NFR-04:** Private routes must be protected using JWT authentication.

- **NFR-05:** API keys, database credentials, and secret keys must be stored in environment variables.
- **NFR-06:** The system should show clear error messages for invalid input or failed requests.
- **NFR-07:** Normal API responses should be reasonably fast.
- **NFR-08:** AI-generated responses may take slightly longer, so the system should show loading feedback.
- **NFR-09:** The project should follow a clean frontend and backend folder structure.
- **NFR-10:** The system should be maintainable so that new features can be added later.
- **NFR-11:** The system should be deployable using Vercel, Render, and MongoDB Atlas.
- **NFR-12:** Sensitive project files such as .env must not be uploaded to GitHub.

User Roles:

Normal User:

A normal user can register, log in, create or update a fitness profile, receive workout and diet recommendations, view exercise guidance, track progress, and receive AI-assisted suggestions.

Responsibilities:

- Register and log in to the system.
- Create and update fitness profile.
- View workout and nutrition recommendations.
- View exercise guidance.
- Track weight and progress.
- View dashboard information.
- Receive AI-assisted feedback and motivation.

Admin:

The admin manages system content, especially exercise-related information and categories.

Responsibilities:

- Log in to the admin panel.
- Add exercise information.

- Update exercise information.
- Delete exercise information.
- Manage exercise categories.
- Keep system content organized.

Use Cases:

Use Case 1: User Registration

Actor: Normal User

Action: User enters name, email, and password.

System Response: The system validates the data, hashes the password, and creates a new user account.

Use Case 2: User Login

Actor: Normal User

Action: User enters email and password.

System Response: The system validates credentials and grants access by generating a JWT token.

Use Case 3: Create Fitness Profile

Actor: Normal User

Action: User enters age, gender, height, weight, activity level, fitness goal, and budget preference.

System Response: The system saves the profile information and calculates basic fitness data such as BMI.

Use Case 4: Get Workout Recommendation

Actor: Normal User

Action: User requests a workout recommendation.

System Response: The system processes the user profile and sends selected data to Gemini AI. The AI returns personalized workout guidance.

Use Case 5: Get Nutrition Recommendation

Actor: Normal User

Action: User requests a nutrition recommendation.

System Response: The system considers user goal, budget, and profile data. Gemini AI returns suitable diet suggestions with local and affordable options.

Use Case 6: Track Progress

Actor: Normal User

Action: User enters current weight or progress information.

System Response: The system stores the progress record and updates the progress history.

Use Case 7: Get Progress Feedback

Actor: Normal User

Action: User requests progress analysis.

System Response: The system checks progress history and sends selected data to Gemini AI. The AI provides feedback, adaptive advice, and motivation.

Use Case 8: View Exercise Library

Actor: Normal User

Action: User opens the exercise library.

System Response: The system displays exercise names, descriptions, difficulty levels, target body parts, and video guidance.

Use Case 9: Manage Exercise Content

Actor: Admin

Action: Admin adds, updates, or deletes exercise information.

System Response: The system updates the exercise library content.

Use Case 10: Manage Exercise Categories

Actor: Admin

Action: Admin adds, updates, or deletes exercise categories.

System Response: The system updates category information for the exercise library.